Name: James Jordon	**Date:** 3/3/2026

Module 13

CompTIA Security+ Exam Objectives

Domain	**Lab**
Security Architecture	13.2, 13.4, 13.5
Security Operations	13.1, 13.2, 13.3, 13.4, 13.5
Security Program Management and Oversight	13.1, 13.2, 13.3, 13.4, 13.5

Website links change frequently. If a link in the following activities no longer works, use a search engine such as Google or Bing to search for the information.

Estimated Time: 50 minutes

Objectives

Linux approaches log files in a different manner from Windows. You should understand and have knowledge of the log file structure in Linux, as many devices are built from Unix and Linux and follow similar structures. Although the varying distributions of Linux may apply log files in different ways, the underlying structure of the files is similar. You should know how to access Linux log files from the command line, as it is common to have only a command-line interface with no graphical user interface on a device running Linux.

After completing this lab, you will be able to do the following:

- Use journalctl to review logs in Linux.
- Locate common log files in Linux.
- View log files in the terminal in Linux.

Certification Objectives

Objectives for the CompTIA Security+ exam:

- 4.8 Explain appropriate incident response activities.

- 4.9 Given a scenario, use data sources to support an investigation.
- 5.2 Explain elements of the risk management process.

Materials Required

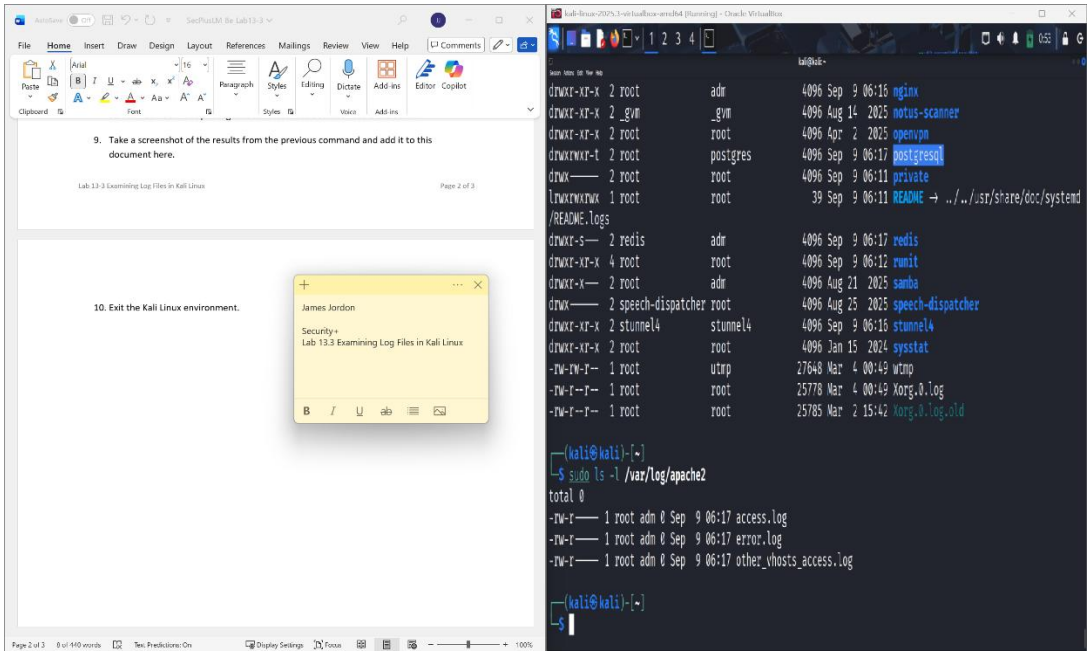
This lab requires the following:

- A computer running the Windows operating system with Internet access
- Kali Virtual Machine (VM)

Activity

In this lab, you examine log files in Kali Linux.

1. Start your Kali VM, log in, and then open a terminal window.
2. Type **sudo journalctl** and press **Enter**.
3. Press **Spacebar** a few times to see more pages of the file.
4. Press **q** to quit.
5. Type **sudo journalctl -n 10** and press **Enter** to display a more manageable number of lines from log files. Notice the logs for a session opening and closing for root. Notice the lines in bold for the command you just ran.
6. Press **q** to quit.
7. Type **ls -l /var/log** and press **Enter**. Notice that root is the owner of most of the log files in Kali Linux. If you run the Kali VM through VMware, you will see VMware log files.
8. Type **sudo ls -l /var/log/apache2** and press **Enter** to view all log files in the Apache2 server. You need root privileges to list these files because root is the owner.
9. Take a screenshot of the results from the previous command and add it to this document here.
10. Exit the Kali Linux environment.


9. Take a screenshot of the results from the previous command and add it to this document here.
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10. Exit the Kali Linux environment.
James Jordon
Security+
Lab 13.3 Examining Log Files in Kali Linux
drwxr-xr-x 2 root adm 4096 Sep 9 06:16 nginx
drwxr-xr-x 2 _gvm _gvm 4096 Aug 14 2025 notus-scanner
drwxr-xr-x 2 root root 4096 Apr 2 2025 openvpn
drwxrwxr-t 2 root postgres 4096 Sep 9 06:17 postgresql
drwx—— 2 root root 4096 Sep 9 06:11 private
lrwxrwxrwx 1 root root 39 Sep 9 06:11 README → ../../usr/share/doc/systemd/README.logs
drwxr-s— 2 redis adm 4096 Sep 9 06:17 redis
drwxr-xr-x 4 root root 4096 Sep 9 06:12 runit
drwxr-x— 2 root adm 4096 Aug 21 2025 samba
drwx—— 2 speech-dispatcher root 4096 Aug 25 2025 speech-dispatcher
drwxr-xr-x 2 stunnel4 stunnel4 4096 Sep 9 06:16 stunnel4
drwxr-xr-x 2 root root 4096 Jan 15 2024 sysstat
-rw-rw-r— 1 root utmp 27648 Mar 4 00:49 wtmp
-rw-r--r— 1 root root 25778 Mar 4 00:49 Xorg.0.log
-rw-r--r— 1 root root 25785 Mar 2 15:42 Xorg.0.log.old
┌──(kali㉿kali)-[~]
└─$ sudo ls -l /var/log/apache2
total 0
-rw-r—— 1 root adm 0 Sep 9 06:17 access.log
-rw-r—— 1 root adm 0 Sep 9 06:17 error.log
-rw-r—— 1 root adm 0 Sep 9 06:17 other_vhosts_access.log
┌──(kali㉿kali)-[~]
└─$